

Lecture 7: Recommender Systems

Recommender Systems

■ Netflix Prize

- The Netflix Prize was an open competition for the best collaborative filtering algorithm to predict user ratings for films, based on previous ratings.
 - The competition was held by Netflix, an online DVD-rental and video streaming service, and the grand prize is US \$1,000,000.
- Business values:
 - Netflix has around 193 million subscribers globally.
 - Netflix claims its AI assisted recommendation system saves the company \$1 billion per year.

Overview

- Collaborative Filtering
 - Memory based approach
 - Model based approach

Recommender Systems

■ What Are Recommender Systems?

- Recommender system is a data mining technique that helps users discover new and relevant items.
- They suggest the most relevant items to buy and, as a result, increase a company's revenue. These suggestions are based on users' behavior and history that contain information on their past preferences.

Recommendation systems changed the way websites communicate with their users. Rather than providing a static experience in which users search for and potentially buy products, recommender systems increase interaction to provide a richer experience.

Types of Recommendations

- Aggregates
 - Trending, Popular, Recent uploads
- Personalized Recommendation
 - Make recommendations based on user personal preference

Data for Recommendation

- **Explicit feedback**

- Explicit feedback are provided by the user. They infer the user's preference. Examples include ratings and reviews.

- **Implicit feedback**

- Implicit feedback are provided when users interact with the item. They infer a user's behavior and are easy to get as users are subconsciously clicking. Examples include clicks, views and purchases.

Data for Recommendation

■ Explicit feedback

	<i>Batman Begins</i>	<i>Alice in Wonderland</i>	<i>Dumb and Dumber</i>	<i>Equilibrium</i>
User A	4	?	3	5
User B	?	5	4	?
User C	5	4	2	?
User D	2	4	?	3
User E	3	4	5	?

■ Implicit feedback

	<i>Item 1</i>	<i>Item 2</i>	<i>Item 3</i>	<i>Item 4</i>
User A	1	?	1	1
User B	?	1	1	?
User C	1	1	1	?
User D	1	1	?	1
User E	1	1	1	?

Types of Recommender Systems

- How to make recommendations?
- The most common recommender system approach is Collaborative Filtering

Collaborative Filtering

Overview of Collaborative Filtering

- The motivation for collaborative filtering comes from the idea that people often get the best recommendations from someone with tastes similar to themselves.
- Collaborative filtering encompasses techniques for matching people with similar interests and making recommendations on this basis.

Common Approach of Collaborative Filtering

There are two common approaches of collaborative filtering:

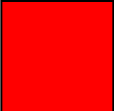
- Memory-based approach
 - **User-User**: measure the similarity between target users and other users
 - **Item-Item**: measure the similarity between the items that target users rates/interacts with and other items
- Model-based approach

Collaborative Filtering

		users											
		1	2	3	4	5	6	7	8	9	10	11	12
movies	1	1		3			5			5		4	
	2			5	4			4			2	1	3
	3	2	4		1	2		3		4	3	5	
	4		2	4		5			4			2	
	5			4	3	4	2					2	5
	6	1		3		3			2			4	
			- unknown rating							- rating between 1 to 5			

Collaborative Filtering

		users											
		1	2	3	4	5	6	7	8	9	10	11	12
movies	1	1		3		?	5			5		4	
	2			5	4			4			2	1	3
	3	2	4		1	2		3		4	3	5	
	4		2	4		5			4			2	
	5			4	3	4	2					2	5
	6	1		3		3			2			4	

 - estimate rating of movie 1 by user 5

User–User Collaborative Filtering

Step 1. identify users rated movie 1

movies	users											
	1	2	3	4	5	6	7	8	9	10	11	12
	1		3		?	5			5		4	
	2		5	4			4			2	1	3
	3	2	4		1	2		3		4	3	5
	4		2	4		5			4		2	
	5			4	3	4	2				2	5
	6	1		3		3			2		4	

User–User Collaborative Filtering

Step 2. Measure the similarity between user 5 and users who rated movie 1.

- 1) Subtract mean rating u_i from each user i
 $u_1 = (1+2+1)/3 = 1.3$
col 1: [-0.3, 0, 0.7, 0, 0, -0.3]
- 2) Compute cosine similarities between cols

	users											
	1	2	3	4	5	6	7	8	9	10	11	12
movies	1	1	3		?	5			5		4	
	2		5	4			4			2	1	3
	3	2	4	1	2		3		4	3	5	
	4		4		5			4			2	
	5		4	3	4	2					2	5
	6	1	3		3			2			4	
	-0.49		0.21			-0.16			0.47		-0.71	

User–User Collaborative Filtering

Step 2. Calculate the predicted rating based on similar users

$$\frac{(-0.8) * 0.21 + 0.5 * 0.47}{0.21 + 0.47} + 3.5 = 3.60$$

		users											
		1	2	3	4	5	6	7	8	9	10	11	12
movies	1	1		3		?	5			5		4	
	2			5	4			4			2	1	3
	3	2	4		1	2		3		4	3	5	
	4		2	4		5			4			2	
	5			4	3	4	2					2	5
	6	1		3		3			2			4	
		-0.49		0.21			-0.16			0.47		-0.71	

Item–Item Collaborative Filtering

- User–user collaborative filtering, while effective, suffers from scalability problems as the user base grows. To extend collaborative filtering to large user bases and facilitate deployment on e-commerce sites, it was necessary to develop more scalable algorithms.
- Item–item collaborative filtering, also called item-based collaborative filtering, takes a major step in this direction and is one of the most widely deployed collaborative filtering techniques today.

Item-Item Collaborative Filtering

		users											
		1	2	3	4	5	6	7	8	9	10	11	12
movies	1	1		3		?	5			5		4	
	2			5	4			4			2	1	3
	3	2	4		1	2		3		4	3	5	
	4		2	4		5			4			2	
	5			4	3	4	2					2	5
	6	1		3		3			2			4	



- estimate rating of movie 1 by user 5

Item–Item Collaborative Filtering

Step 1. identify movies rated by user 5

		users											
		1	2	3	4	5	6	7	8	9	10	11	12
movies	1	1		3		?	5			5		4	
	2			5	4			4			2	1	3
	3	2	4		1	2		3		4	3	5	
	4		2	4		5			4			2	
	5			4	3	4	2					2	5
	6	1		3		3			2			4	

Item–Item Collaborative Filtering

Step 2. Measure the similarity between movie 1 and other movies rated by user 5.

- 1) Subtract mean rating m_i from each movie i
 $m_1 = (1+3+5+5+4)/5 = 3.6$
row 1: $[-2.6, 0, -0.6, 0, 0, 1.4, 0, 0, 1.4, 0, 0.4, 0]$
- 2) Compute cosine similarities between rows

		users													
		1	2	3	4	5	6	7	8	9	10	11	12		
movies	1	1		3		?	5			5		4			
	2			5	4			4			2	1	3		
	3	2	4		1	2		3		4	3	5		0.41	
	4		2	4		5			4			2		-0.10	
	5			4	3	4	2					2	5	-0.31	
	6	1		3		3			2			4		0.59	

Item–Item Collaborative Filtering

Step 3. Calculate the predicted rating based on similar items.

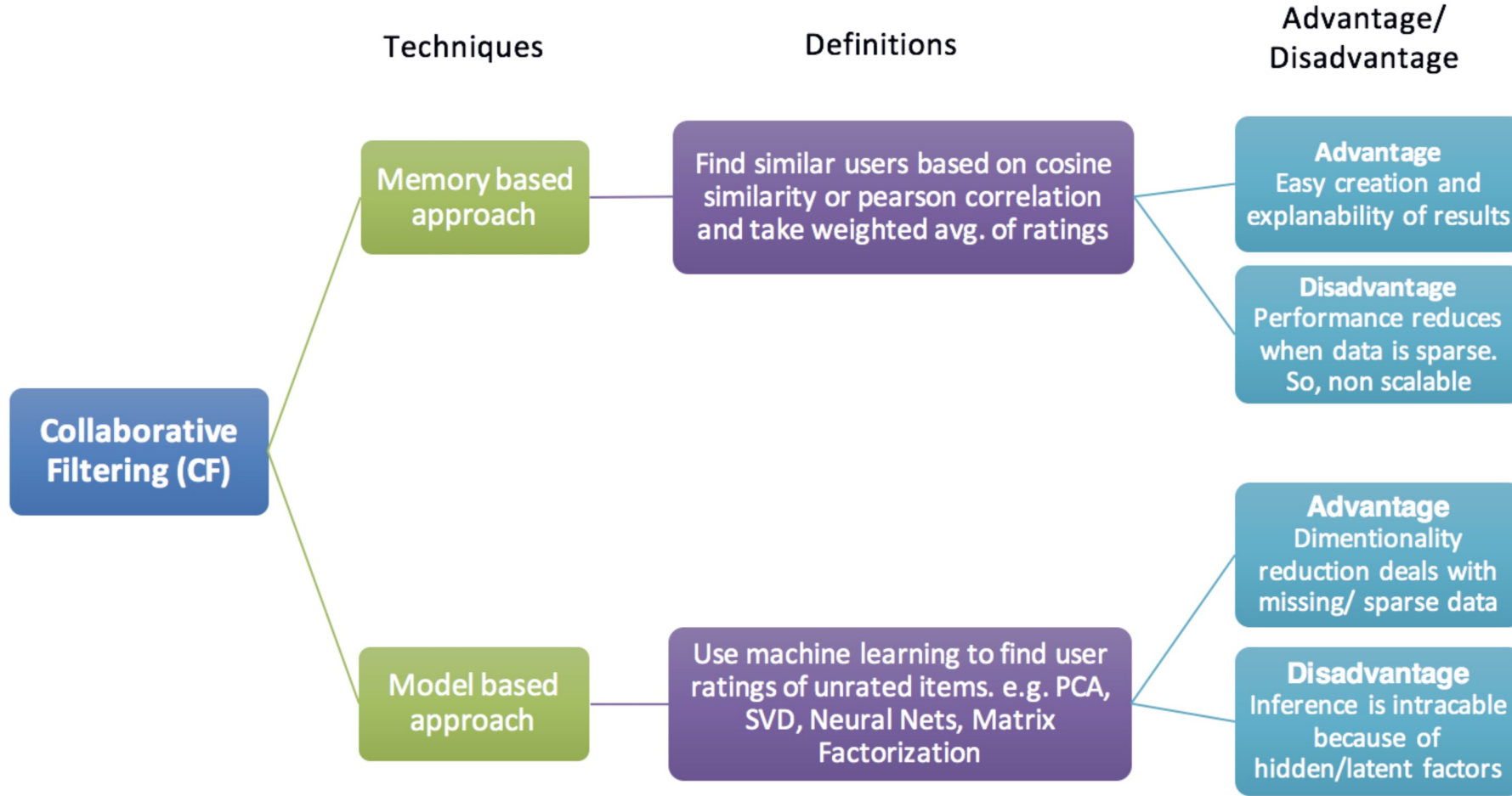
$$(0.41 \cdot 2 + 0.59 \cdot 3) / (0.41 + 0.59) = 2.6$$

	1	2	3	4	5	6	7	8	9	10	11	12	
1	1		3		?	5			5		4		
2			5	4			4			2	1	3	
3	2	4		1	2		3		4	3	5		0.41
4		2	4		5			4			2		-0.10
5			4	3	4	2					2	5	-0.31
6	1		3		3			2			4		0.59

Pros/Cons of Collaborative Filtering

- **+ Works for any kind of item**
- **- Cold Start:**
 - Need enough users in the system to find a match
- **- Sparsity:**
 - The user/ratings matrix is sparse
 - Hard to find users that have rated the same items
- **- Popularity bias:**
 - Cannot recommend items to someone with unique taste
 - Tends to recommend popular items

Common Approach of Collaborative Filtering



Steps to Build A Neural Collaborative Filtering

- Step 1: MovieLense Data
- Step 2: Data Preprocessing
- Step 3: Build a Neural Collaborative Filtering on Training Data
- Step 4: Evaluate Neural Collaborative Filtering Performance

Step 1: MovieLense Data

- We will use the MovieLense 1M ratings data (downloaded from <http://www.grouplens.org/>), which contains around 1,000,000 ratings (1-5) from 6,000 users on 4,000 movies.
 - User description: user id, gender, age, occupation, zip code
 - Movie information: movie id, title, genre
 - Ratings: user id, movie, id, rating, timestamp

Step 1: MovieLense Data

■ User description

	user_id	gender	age	occupation	zip_code
0	1	F	1	10	48067
1	2	M	56	16	70072
2	3	M	25	15	55117
3	4	M	45	7	02460
4	5	M	25	20	55455
...	...	...	...	...	...
6035	6036	F	25	15	32603
6036	6037	F	45	1	76006
6037	6038	F	56	1	14706
6038	6039	F	45	0	01060
6039	6040	M	25	6	11106

6040 rows × 5 columns

Step 1: MovieLense Data

■ Movie information

	movie_id	title	genre
0	1	Toy Story (1995)	Animation Children's Comedy
1	2	Jumanji (1995)	Adventure Children's Fantasy
2	3	Grumpier Old Men (1995)	Comedy Romance
3	4	Waiting to Exhale (1995)	Comedy Drama
4	5	Father of the Bride Part II (1995)	Comedy
...	...	...	...
3878	3948	Meet the Parents (2000)	Comedy
3879	3949	Requiem for a Dream (2000)	Drama
3880	3950	Tigerland (2000)	Drama
3881	3951	Two Family House (2000)	Drama
3882	3952	Contender, The (2000)	Drama Thriller

3883 rows × 3 columns

Step 1: MovieLense Data

■ Ratings

	user_id	movie_id	rating	timestamp
0	1	1193	5	978300760
1	1	661	3	978302109
2	1	914	3	978301968
3	1	3408	4	978300275
4	1	2355	5	978824291
...	...	...	...	...
1000204	6040	1091	1	956716541
1000205	6040	1094	5	956704887
1000206	6040	562	5	956704746
1000207	6040	1096	4	956715648
1000208	6040	1097	4	956715569

1000209 rows × 4 columns

Step 1: MovieLense Data

■ Merged data

	movie_id	title	genre	user_id	rating	timestamp	gender	age	occupation	zip_code
0	1	Toy Story (1995)	Animation Children's Comedy	1	5	978824268	F	1	10	48067
1	48	Pocahontas (1995)	Animation Children's Musical Romance	1	5	978824351	F	1	10	48067
2	150	Apollo 13 (1995)	Drama	1	5	978301777	F	1	10	48067
3	260	Star Wars: Episode IV - A New Hope (1977)	Action Adventure Fantasy Sci-Fi	1	4	978300760	F	1	10	48067
4	527	Schindler's List (1993)	Drama War	1	5	978824195	F	1	10	48067
...	...	...	...	...	...	...	...	...	...	...
1000204	3513	Rules of Engagement (2000)	Drama Thriller	5727	4	958489970	M	25	4	92843
1000205	3535	American Psycho (2000)	Comedy Horror Thriller	5727	2	958489970	M	25	4	92843
1000206	3536	Keeping the Faith (2000)	Comedy Romance	5727	5	958489902	M	25	4	92843
1000207	3555	U-571 (2000)	Action Thriller	5727	3	958490699	M	25	4	92843
1000208	3578	Gladiator (2000)	Action Drama	5727	5	958490171	M	25	4	92843

1000209 rows x 10 columns

Step 2: Data Preprocessing

■ Encode movie_id and user_id

	user_id	movie_id	rating	timestamp
0	0	1104	5	978300760
1	0	639	3	978302109
2	0	853	3	978301968
3	0	3177	4	978300275
4	0	2162	5	978824291
...	...	...	...	...
1000204	6039	1019	1	956716541
1000205	6039	1022	5	956704887
1000206	6039	548	5	956704746
1000207	6039	1024	4	956715648
1000208	6039	1025	4	956715569

1000209 rows x 4 columns

Step 2: Data Preprocessing

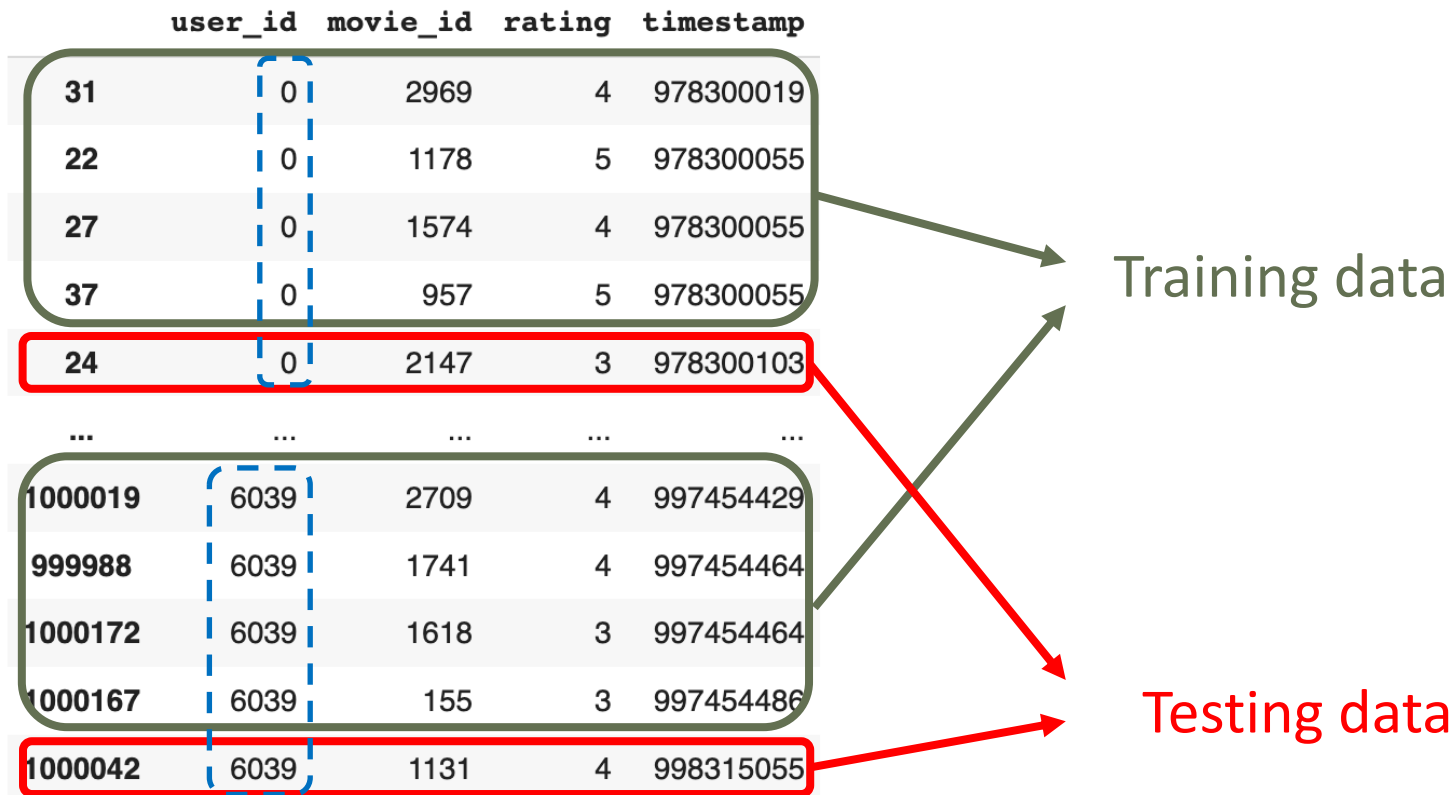
- Sort data based on 'user_id' and 'timestamp'

	user_id	movie_id	rating	timestamp
31	0	2969	4	978300019
22	0	1178	5	978300055
27	0	1574	4	978300055
37	0	957	5	978300055
24	0	2147	3	978300103
...	...	...	...	...
1000019	6039	2709	4	997454429
999988	6039	1741	4	997454464
1000172	6039	1618	3	997454464
1000167	6039	155	3	997454486
1000042	6039	1131	4	998315055

1000209 rows × 4 columns

Step 2: Data Preprocessing

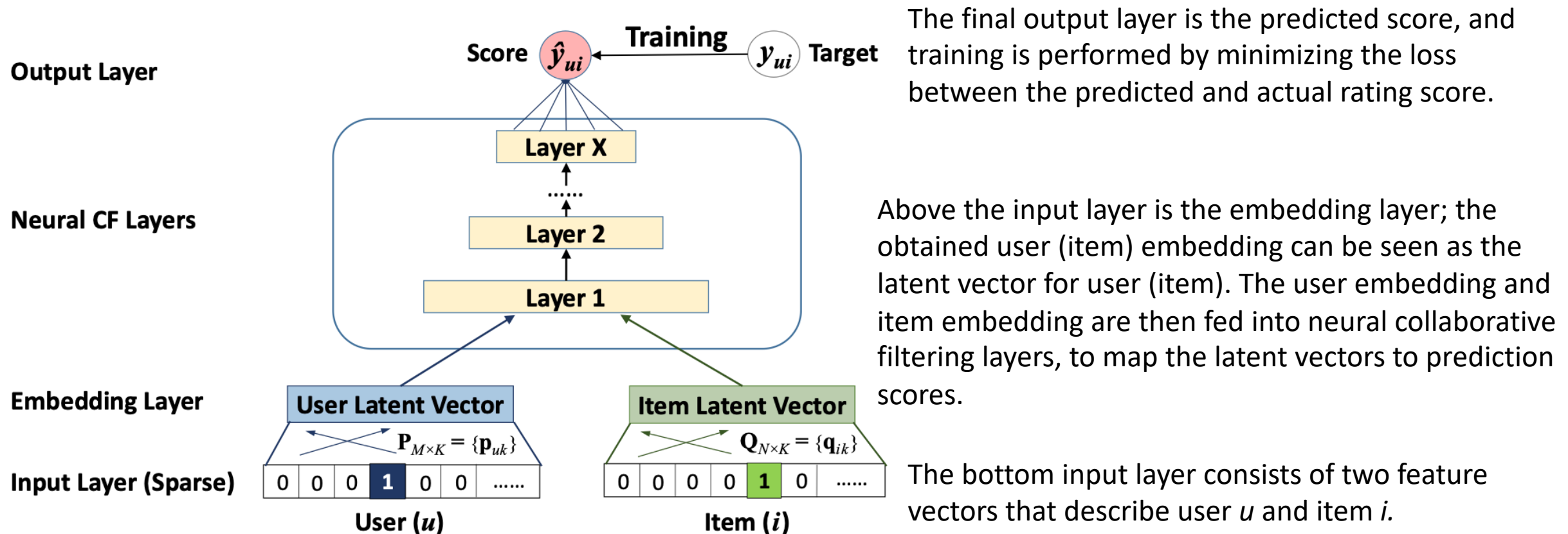
■ Partition the data



1000209 rows x 4 columns

Step 3: Neural Collaborative Filtering

- He, X., et al. Neural collaborative filtering. *Proceedings of the 26th international conference on world wide web*. 2017.



Step 3: Neural Collaborative Filtering

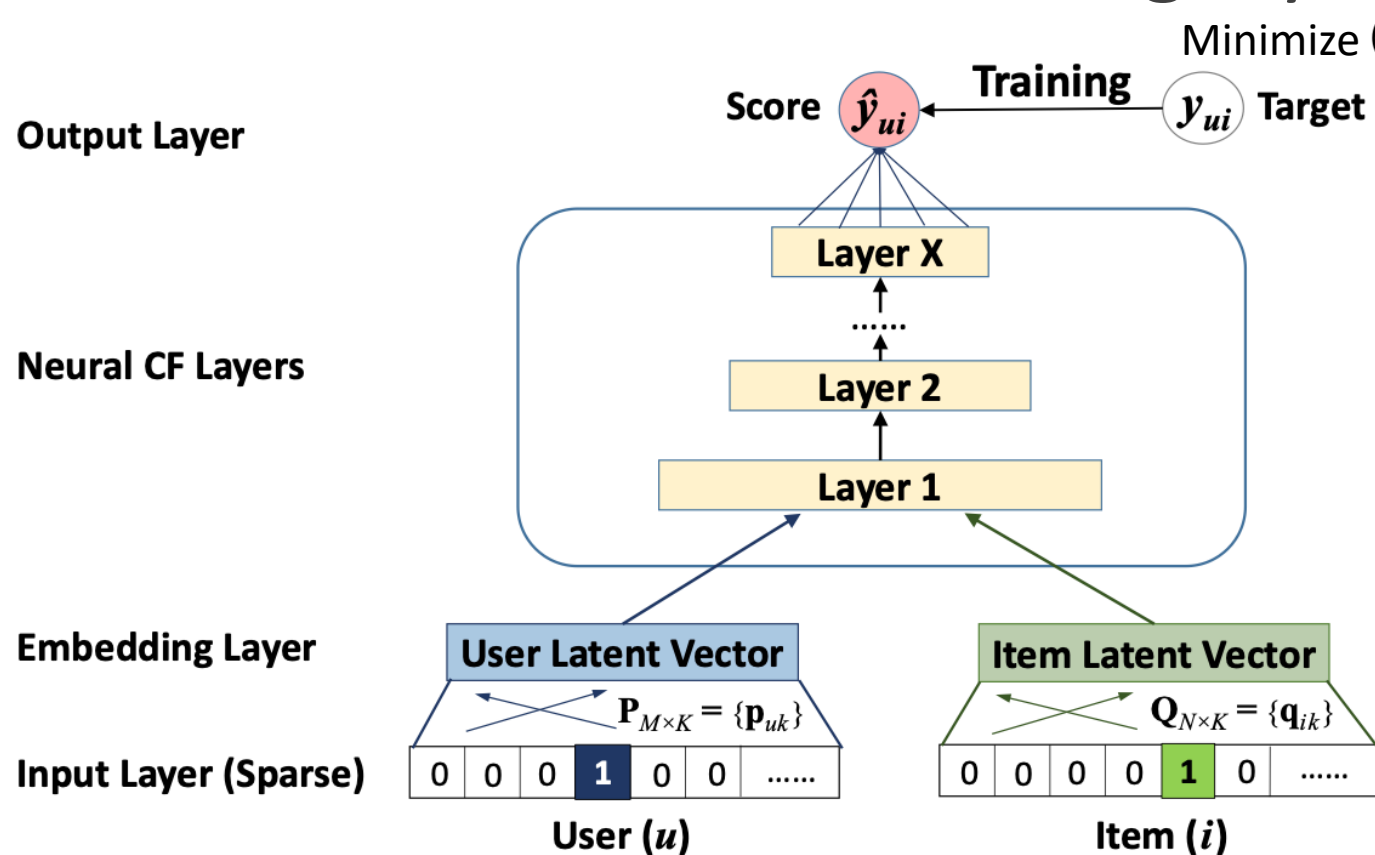
- User/item embedding
 - Embedding maps users/items to low-dimensional vectors in a way that similar users/items are close to each other.

user	Embedding vector							
1	0.6	0.9	0.1	0.4	-0.7	-0.3	-0.2	0.4
2	0.5	0.8	0.2	0.3	-0.9	-0.1	-0.1	0.4
3	-0.4	-0.5	0.5	0.6	0.8	-0.8	0.2	0.5

- Embedding transform user/item id into numbers (input must be numeric for neural networks)
- Embedding transform learns similarities among users/items

Step 3: Neural Collaborative Filtering

■ Neural collaborative filtering layers



Design of network structure follows a tower pattern, where the bottom layer is the widest and each successive layer has a smaller number of neurons. The premise is that by using a small number of hidden units for higher layers, they can learn more abstractive features of data.

Step 4: Evaluate Neural Collaborative Filtering Performance

■ Predict ratings

	user_id	movie_id	rating	timestamp
31	0	2969	4	978300019
22	0	1178	5	978300055
27	0	1574	4	978300055
37	0	957	5	978300055
24	0	2147		978300103
...	...	...	...	...
1000019	6039	2709	4	997454429
999988	6039	1741	4	997454464
1000172	6039	1618	3	997454464
1000167	6039	155	3	997454486
1000042	6039	1131		998315055

1000209 rows x 4 columns

Predict ratings

Testing data

Step 4: Evaluate Neural Collaborative Filtering Performance

■ Mean Absolute Error (MAE)

- The **mean absolute error (MAE)** is a quantity used to measure how close forecasts or predictions are to the eventual outcomes.
- This is known as a scale-dependent accuracy measure and therefore cannot be used to make comparisons between series using different scales

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|$$

■ Root Mean Squared Error (RMSE)

- **RMSE** gives a relatively high weight to large errors. This means the RMSE should be more useful when large errors are particularly undesirable.

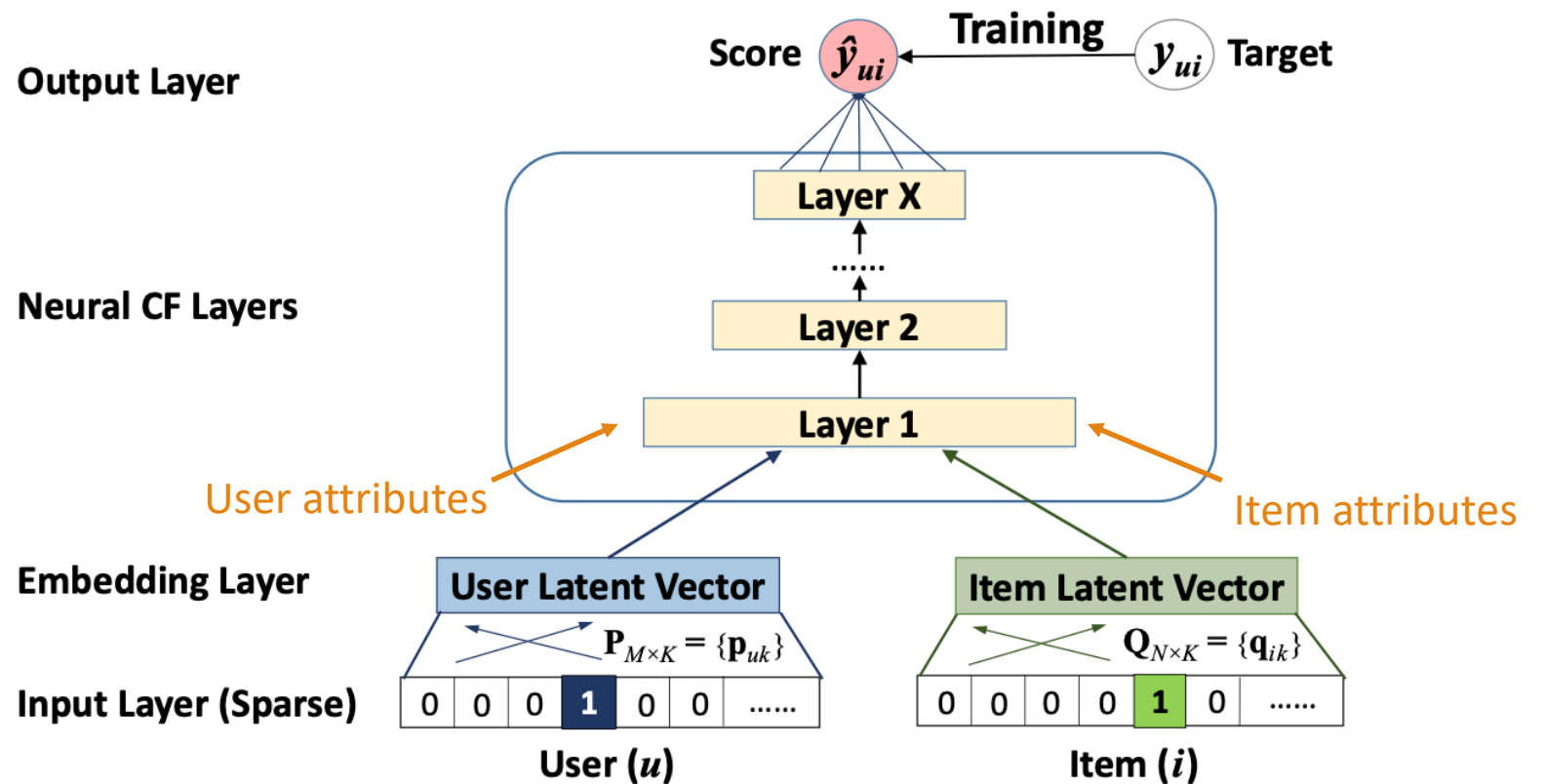
$$\text{RMSE} = \sqrt{\frac{\sum_{t=1}^n (y_t - \hat{y}_t)^2}{n}}$$

Training MAE and RMSE: 0.6683394005368883 0.8520005597969494

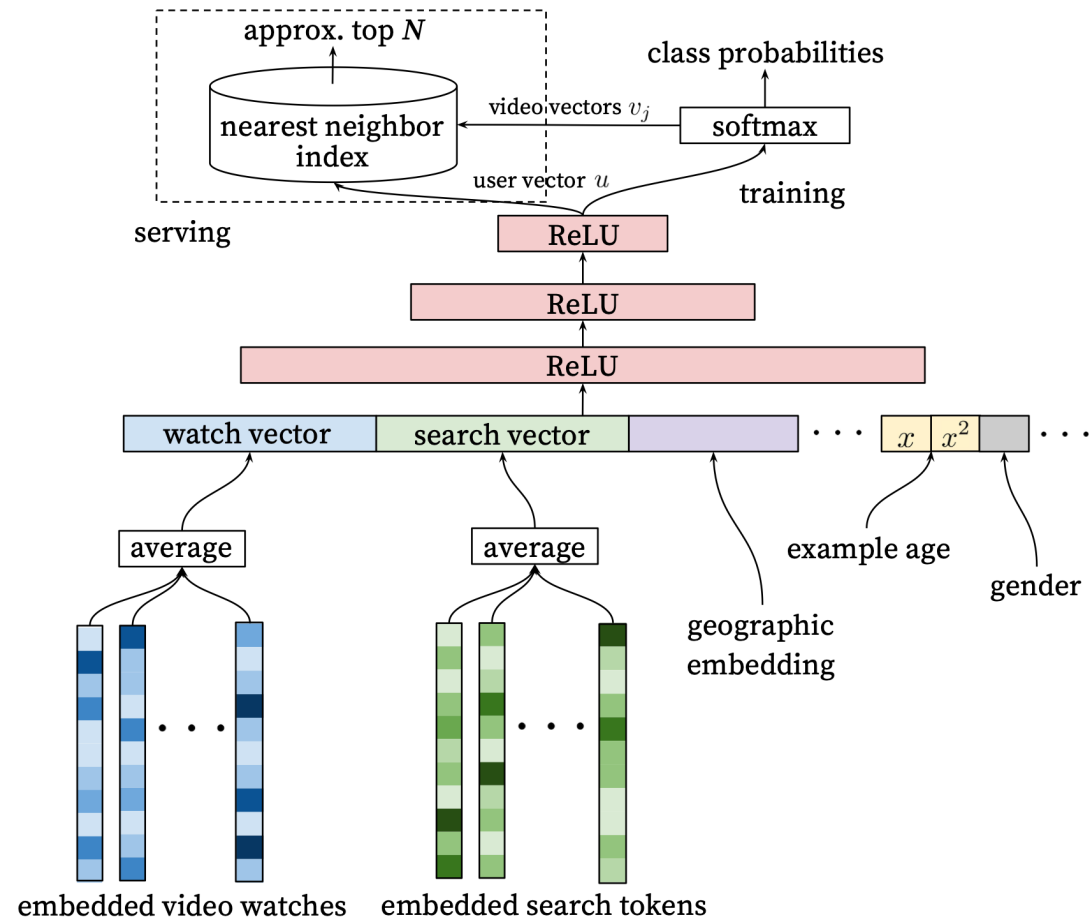
Testing MAE and RMSE: 0.7322674097999042 0.9388386549274153

Collaborative Filtering

- Extend Neural collaborative filtering with additional user and item attributes.



Collaborative Filtering



Covington, Paul, Jay Adams, and Emre Sargin. "Deep neural networks for youtube recommendations." *Proceedings of the 10th ACM conference on recommender systems*. 2016.